$$
\begin{tikzcd}
& \textbf{Hidden Constraint} \\ \text{(The Syzygy)} \arrow[d, dotted, no head] & \textbf{Instructions} \\ \text{(Free Module)} \arrow[d, dotted, no head] & \textbf{Visible Pieces} \\ \text{(The Ideal)} \arrow[d, dotted, no head] & \\
0 \arrow[r] & R \arrow[r, "\psi = \begin{pmatrix} -y^2 \\ x^2 \end{pmatrix}"] & R^2 \arrow[r, "\phi = \begin{pmatrix} x^2 & y^2 \end{pmatrix}"] & (x^2, y^2) \arrow[r] & 0 \\
& c \arrow[r, "\cdot\psi"'] & \begin{pmatrix} -cy^2 \\ cx^2 \end{pmatrix} \arrow[r, "\cdot\phi"'] & 0 &
\end{tikzcd}
$$

Geometric View of Complete Intersection: $I = (x^2, y^2)$

- **Generators:** x^2 and y^2 vanish exactly at the origin $(0, 0)$.
- **Syzygy:** The unique Koszul relation $(-y^2)x^2 + (x^2)y^2 = 0$.
- **Resolution:** Terminates in a single step (codimension 2). No higher syzygies!